Database Normalization:

First Normal Form (1NF)

Definition: A relation is in 1NF if it contains only atomic (indivisible) values; each column contains unique and single values, and each record (row) is unique.

Example: Consider a table storing customer orders:

OrderID	CustomerName	ItemsOrdered
1	John Doe	Pen, Notebook
2	Jane Smith	Pencil, Eraser

Normalization to 1NF:

OrderID	CustomerName	Item
1	John Doe	Pen
1	John Doe	Notebook
2	Jane Smith	Pencil
2	Jane Smith	Eraser

Second Normal Form (2NF)

Definition: A relation is in 2NF if it is in 1NF and all non-key attributes are fully functionally dependent on the entire primary key.

Example: Consider a table with composite primary key (OrderID, Item):

OrderID	Item	CustomerName	OrderDate
1	Pen	John Doe	2023-01-01
1	Notebook	John Doe	2023-01-01
2	Pencil	Jane Smith	2023-01-02
2	Eraser	Jane Smith	2023-01-02

Normalization to 2NF: Separate into two tables to ensure no partial dependency:

Orders Table:

OrderID	CustomerName	OrderDate
1	John Doe	2023-01-01

OrderID	CustomerName	OrderDate
2	Jane Smith	2023-01-02

OrderItems Table:

OrderID	Item
1	Pen
1	Notebook
2	Pencil
2	Eraser

Third Normal Form (3NF)

Definition: A relation is in 3NF if it is in 2NF and all the attributes are functionally dependent only on the primary key.

Example: Consider the Orders table:

OrderID	CustomerName	OrderDate	CustomerAddress
1	John Doe	2023-01-01	123 Elm St
2	Jane Smith	2023-01-02	456 Oak St

Normalization to 3NF: Separate into two tables to remove transitive dependency:

Orders Table:

OrderID	CustomerID	OrderDate
1	1	2023-01-01
2	2	2023-01-02

Customers Table:

CustomerID	CustomerName	CustomerAddress
1	John Doe	123 Elm St
2	Jane Smith	456 Oak St

Boyce-Codd Normal Form (BCNF)

Definition: A relation is in BCNF if it is in 3NF and for every non-trivial functional dependency, the left side is a superkey.

Example: Consider a table with the following structure:

StudentID	CourseID	Instructor
1	A	Dr. Smith
2	B	Dr. Jones

If a course is always taught by the same instructor, the table is not in BCNF:

Normalization to BCNF: Separate into two tables:

Courses Table:

CourseID	Instructor
A	Dr. Smith
B	Dr. Jones

StudentCourses Table:

StudentID	CourseID
1	A
2	B

Fourth Normal Form (4NF)

Definition: A relation is in 4NF if it is in BCNF and has no multi-valued dependencies.

Example: Consider a table with multi-valued dependencies:

StudentID	CourseID	Hobby
1	A	Reading
1	A	Swimming
1	B	Reading
1	B	Swimming

Normalization to 4NF: Separate into two tables to ensure no multi-valued dependencies:

StudentCourses Table:

StudentID	CourseID
1	A
1	B

StudentHobbies Table:

StudentID	Hobby
1	Reading
1	Swimming

Fifth Normal Form (5NF)

Definition: A relation is in 5NF if it is in 4NF and every join dependency is implied by the candidate keys.

Example: Consider a table where the data can be decomposed into multiple tables without loss of information:

ProjectID	EmployeeID	Role
1	1	Developer
1	2	Manager
2	1	Architect

Normalization to 5NF: Separate into multiple tables:

Projects Table:

ProjectID	Role
1	Developer
1	Manager
2	Architect

EmployeeProjects Table:

ProjectID	EmployeeID
1	1
1	2
2	1

Sixth Normal Form (6NF)

Definition: A relation is in 6NF if it is in 5NF and it manages time-variant data (temporal data).

Example: Consider a table that stores historical data:

EmployeeID	Department	StartDate	EndDate
1	Sales	2023-01-01	2023-06-30
1	IT	2023-07-01	NULL

Normalization to 6NF: Separate into tables that isolate changes over time:

EmployeeDepartments Table:

EmployeeID	Department	StartDate	EndDate
1	Sales	2023-01-01	2023-06-30
1	IT	2023-07-01	NULL

Non-Temporal Data Table:

EmployeeID	OtherAttributes
1	...

Seventh Normal Form (7NF) is not widely recognized or used in practice.